

The Final Project

Week 10 — Tuesday Lecture

The job

You have been hired as a **contractor** to fly a drone mission for a client: a city, a contractor, a property owner.

1. Find a civil or construction engineering question drone data can answer
2. Plan the project and the flight
3. Fly it
4. Process the data
5. Present it in class
6. Write a one-page technical memo

Anyone can make an orthomosaic. The engineering part is saying **who needs the data, why you flew it that way, and what it can and cannot support.**

Groups and grade

Groups of 2. Both partners get the same score, except attendance and presenting.

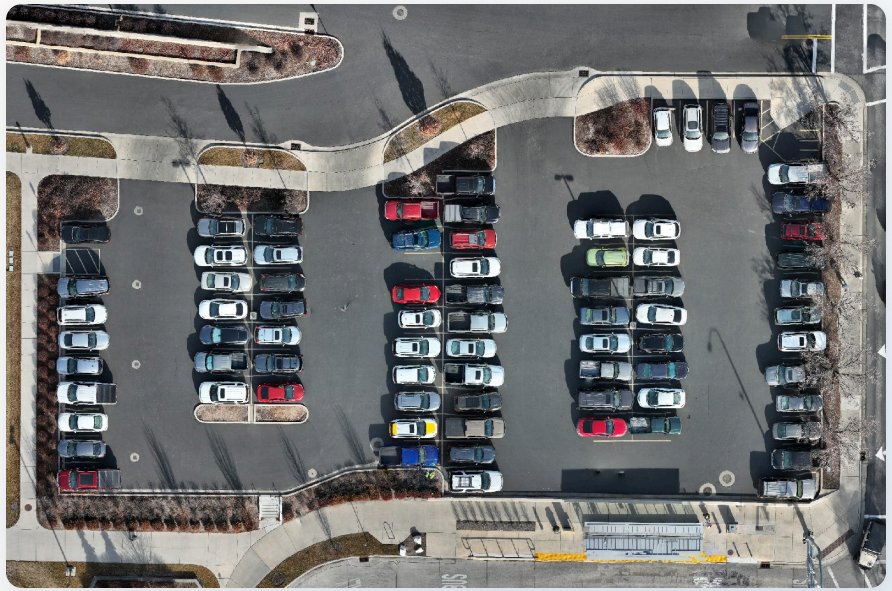
#	Turn-in	Weight	Due
1	Project Proposal	2.5%	**Thu, Nov 5, 11:59 PM**
2	Flight and Data Processing	5%	One week after your flight
3	Presentation	2.5%	Slides Mon, Dec 7; present Dec 8 or 10
4	Technical Memo	5%	Thu, Dec 10
	Total	**15%**	

Final project work cannot be turned in late.

What could you fly?

- Pavement condition on a road or lot: where are repairs needed?
- Room for more parking stalls, a trail, or a new structure
- Roof, field, or paved area, for a runoff calculation
- Volume of a stockpile, or the shape of a slope
- Progress on a construction site

More, with what went wrong for past groups: **Final Project → Example Projects**



Weak or strong?

"We will make a 3D model of the Wilkinson Center."

vs.

"Facilities wants to know how much of the EB east lot needs resurfacing. We will map it and measure the cracked area."

Turn-in 1: the Proposal

Due **Thursday, Nov 5 at 11:59 PM**. Thursday's lab is work time.

1. **Project and purpose:** who is the client, what will they decide?
2. **Site:** map, why it fits, hazards
3. **Airspace, permissions, and safety**
4. **Time budget:** the whole trip in 50 minutes
5. **Flight plan:** built in the class planner, with a reason for every setting, and a backup drone
6. **Products and roles:** pilot, visual observer, data

Upload: proposal PDF + KMZ + Mission Log + permissions. **No approved proposal, no flight.**

Permissions are your job

- **Airspace:** most of campus is in Provo's Class D. If you need **LAANC**, fill it out **with a TA** (it is filed under their Part 107 certificate).
- **Property:** private land, active construction, and some campus areas need the **owner's OK**, in writing. An email is fine.
- **People and traffic:** how will you stay off people and moving cars?

Turn in with the proposal: the LAANC request, and the owner's OK **or** who you contacted, how, and when.

You must have the OK **before you fly**. Proof is graded in turn-in 2.

The 50-minute trip

Step	Minutes
Check out drone at EB B122	
Travel to site	
Set up + pre-flight checklist	
Flight time (from your plan)	
Extra photos or measurements	
Post-flight + pack up	
Travel back to EB B122	
Total	**≤ 50**

Two trips each flight lab: **8:00–8:50** and **9:00–9:50**.

The drone goes out again with the next trip.

Plan for 40, not 50. Wind, a pedestrian, or a battery swap will use the rest.

Flight days

	8:00–8:50	9:00–9:50
Thu, Nov 12	3 groups	2 groups
Thu, Nov 19	2 groups	2 groups

One of each drone per trip, so you sign up for a **drone on a trip**:

- **Mini:** regular photos. Use it unless you need the others.
- **Thermal** (Autel): roofs, pavement, moisture
- **Multispectral:** vegetation health (NDVI)

Sign-up sheet is linked on the Project Overview. First come, first served.

Flight day

Be at EB B122 by the start of your trip. It leaves on time.

- A partner who is not there is left behind: **0 for attendance**.
- If **neither** partner is there: the group gets 0 and must find a new time with a TA.
- Run the pre-flight and post-flight checklists. Write down anything that changes.
- Start processing **that same week**. Bentley runs for hours on shared machines.

Turn-in 2: Flight and Data Processing

Due **one week after you fly**. Most of the points are for **output**.

Criterion	Points
Mission plan and permissions	5
Flight attendance *(per student)*	10
Processed products: two or more, with a measurement	25
Raw data shared: every original photo	10
Total	**50**

Products and screenshots go to Learning Suite; raw photos go to the class shared folder.

Turn-ins 3 and 4

Presentation: 12 minutes + 5 minutes of questions

- Slides due **Mon, Dec 7**. Present **Tue Dec 8** (3 groups) or **Thu Dec 10** (6 groups)
- Both partners present. A student who does not present gets **0**.

Technical Memo: due **Thu, Dec 10**

- Written to a project manager who was not there
- **One page of writing**, plus figures
- See **Example Technical Memo** on the site

Check one number against something measured another way. Not required, but it is the strongest evidence your data can be trusted.

Attendance, Weeks 10–15

Taken	Only if your group flies	Optional	No class
Nov 3, 5, 10, 17	Nov 12	Nov 24 work day	Nov 26
Dec 1, 8, 10	Nov 19	Dec 3 work day	

Now

Find a partner.

Pick a question and a site.

Check the airspace.

Proposal due **Thursday, 11:59 PM**.